# MUHAMMAD AHMED

## CS Sophomore | Aspiring AI & Software Engineer

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### SUMMARY

CS sophomore at FAST-NUCES Lahore with a strong CGPA of 3.68 and hands-on experience in building complex systems using Python, C /C++, and Assembly. I am quite invested in concurrent design, SOLID principles, and data-driven thinking, and I am eager to leverage my skills in backend, systems, or AI engineering roles

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### EXPERIENCE

**Event Officer SOFTEC '25, FAST-NUCES Lahore**

03/2024 - 04/2024 Lahore, Pakistan

Organized annual tech event at FAST-NUCES Managed end-to-end event flow, handling planning and on-ground execution to ensure smooth coordination across sessions.

**Private Tutor Private Tutor**

02/2025 - Present Lahore, Pakistan

Private tutoring for O-Level students Delivering one-on-one tutoring to an O-Level student, strengthening subject fundamentals and exam technique.

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### EDUCATION

- **Intermediate, Pre-Engineering**
  Aug 2022-May 2024
  Government College University, Lahore

- **Bachelor of Computer Science**
  Aug 2024-Present
  FAST-National University of Computer and Emerging Science (Dean's List Holder)
  CGPA: 3.68 / 4.0

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### SKILLS

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|---|---|---|---|---|
| Bootstrap | SQL | OOP | Express.js | GSAP |
| Data Structures | MySQL | Python | JavaScript | C/C++ |
| HTML | Git | Node.js | GitHub | |
| MSSQL Server | Modularization | | Assembly Language | |

### KEY ACHIEVEMENTS

**Event Coordination Success**

- Achieved 98% satisfaction in event feedback at SOFTEC '25.

**Improved Academic Performance**

- Increased student's grades by 30% after two months of private tutoring.

**High-Throughput Data Pipeline**

- Designed real-time pipeline handling multiple data points per second.

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### PROJECTS

**Realtime Dengue Outbreak Predictor** **Ongoing**

- Leading a team to develop a full-stack dengue predictor for Lahore, integrating WHO surveillance and NASA climate data into an ML pipeline using XGBoost and Random Forest. The system maps geographical hotspots via KMeans clustering.

**HireAtlas Career Management** **Ongoing**

- Developing a unified web platform with a dual-role account architecture to streamline job posting and application tracking. The system utilizes **a relational schema** to manage complex user profiles, experience, and resumes within a centralized digital marketplace.

**Concurrent Data Pipeline** **03/2026**

- Architected a concurrent Python pipeline using SOLID principles and Multiprocessing for high-throughput streams, integrating PBKDF2 verification and Observer-based telemetry to ensure data integrity across dynamic schemas.

**Real-Time GDP Analyzer** **02/2026**

- Analyzes GDP data through a modular design Modular Python framework for economic research using SOLID principles and Dependency Inversion, analyzing multi dimensional GDP datasets via extensible plugin system

**Rush Hour** **11/2025**

- Assembly language-based racing game Road racing game in 8088 Assembly using timer/keyboard interrupts, randomized traffic generation, and collision detection

**Space Wars** **12/2024**

- C++ based 2D arcade shooter game 2D arcade shooter with smooth rendering, responsive controls, collision detection, and power-up